

MAT 362 HW04

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Problem. 2.3.6 Prove that the definition of a differentiable map between surfaces does not depend on the parametrization chosen.

Solution. Let $f: S_1 \rightarrow S_2$ be differentiable at $p \in S_1$ with respect to parametrizations $\mathbf{x}_1: U_1 \subseteq \mathbb{R}^2 \rightarrow S_1$ and $\mathbf{x}_2: U_2 \subseteq \mathbb{R}^2 \rightarrow S_2$ (where $p \in \mathbf{x}_1(U_1)$, $f(p) \in \mathbf{x}_2(U_2)$); that is,

$$\mathbf{x}_2^{-1} \circ f \circ \mathbf{x}_1: U_1 \rightarrow U_2$$

is differentiable at $\mathbf{x}_1^{-1}(p)$.

Then, given parametrizations $\bar{\mathbf{x}}_1: V_1 \subseteq \mathbb{R}^2 \rightarrow S_1$ and $\bar{\mathbf{x}}_2: V_2 \subseteq \mathbb{R}^2 \rightarrow S_2$ such that $p \in \bar{\mathbf{x}}_1(V_1)$ and $f(p) \in \bar{\mathbf{x}}_2(V_2)$, we want to show that

$$\bar{\mathbf{x}}_2^{-1} \circ f \circ \bar{\mathbf{x}}_1: V_1 \rightarrow V_2$$

is differentiable at $\bar{\mathbf{x}}_1^{-1}(p)$. This follows from the “change of parameters” maps

$$\bar{\mathbf{x}}_1^{-1} \circ \mathbf{x}_1: \mathbf{x}_1^{-1}(W_1) \rightarrow \bar{\mathbf{x}}_1^{-1}(W_1), \quad \bar{\mathbf{x}}_2^{-1} \circ \mathbf{x}_2: \mathbf{x}_2^{-1}(W_2) \rightarrow \bar{\mathbf{x}}_2^{-1}(W_2)$$

being diffeomorphisms (where W_1 is $\mathbf{x}_1(U_1) \cap \bar{\mathbf{x}}_1(V_1)$ and W_2 is $\mathbf{x}_2(U_2) \cap \bar{\mathbf{x}}_2(V_2)$), as we can write

$$\bar{\mathbf{x}}_2^{-1} \circ f \circ \bar{\mathbf{x}}_1 = (\bar{\mathbf{x}}_2^{-1} \circ \mathbf{x}_2) \circ (\mathbf{x}_2^{-1} \circ f \circ \mathbf{x}_1) \circ (\mathbf{x}_1^{-1} \circ \bar{\mathbf{x}}_1^{-1})$$

as a composition of differentiable functions. □

Problem. 2.3.7. Prove that the relation “ S_1 is diffeomorphic to S_2 ” is an equivalence relation on the set of regular surfaces.

Solution. The relation is reflexive because the identity map is trivially a diffeomorphism $S \rightarrow S$ for all regular surfaces S .

The relation is symmetric by definition: if there exists a diffeomorphism $f: S_1 \rightarrow S_2$ that is differentiable and has differentiable inverse, then $f^{-1}: S_2 \rightarrow S_1$ is differentiable with differentiable inverse.

The relation is transitive because given surfaces S_1, S_2, S_3 with diffeomorphisms $f: S_1 \rightarrow S_2$ and $g: S_2 \rightarrow S_3$, the composition $g \circ f: S_1 \rightarrow S_3$ is also a diffeomorphism: $g \circ f$ and its inverse $f^{-1} \circ g^{-1}$ are both differentiable because compositions of differentiable functions are differentiable.

Having checked all three conditions for a relation to be an equivalence relation, we can conclude that being diffeomorphic is an equivalence relation on the set of regular surfaces. □

Problem. 2.3.11. Prove that the rotations of a surface of revolution S about its axis are diffeomorphisms of S .

Solution. Suppose WLOG that the z -axis is the rotation axis of S . Then counterclockwise rotation about this axis by some fixed θ is the linear map given by the matrix

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

in the standard basis. Linear maps are differentiable, and since R_θ is invertible, it has an inverse which is of course also differentiable. Thus rotations about an axis are diffeomorphisms of $\mathbb{R}^3$. Then, since S is a surface of revolution whose axis of rotation is the z -axis, the restriction $R_\theta|_S$ is a bijection on S and retains its differentiability properties (cf. Example 3 in the book), so rotations $R_\theta|_S$ are a diffeomorphisms of S . □

Problem. 2.3.15. Let C be a regular curve and let $\alpha: I \subseteq \mathbb{R} \rightarrow C$, $\beta: J \subseteq \mathbb{R} \rightarrow C$ be two parametrizations of C in a neighborhood of $p \in \alpha(I) \cap \beta(J) = W$. Let

$$h = \alpha^{-1} \circ \beta: \beta^{-1}(W) \rightarrow \alpha^{-1}(W)$$

be the change of parameters. Prove that

- (a) h is a diffeomorphism.
- (b) The absolute value of the arc length of C in W does not depend on which parametrization is chosen to define it; that is,

$$\left| \int_{t_0}^t |\alpha'(t)| dt \right| = \left| \int_{\tau_0}^{\tau} |\beta'(\tau)| d\tau \right|, \quad t = h(\tau), \quad t \in I, \quad \tau \in J$$

Solution. (a) First, h must be a homeomorphism because it is a composition of homeomorphisms α^{-1} and β . Let $r \in \beta^{-1}(W)$ and set $q = h(r)$. Since $\alpha(t) = (x(t), y(t), z(t))$ is a parametrization, we can assume, by renaming the axes if necessary, that $x'(q) \neq 0$.

We extend α to a map $F: I \times \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by

$$F(t, u, v) = (x(t), y(t) + u, z(t) + v).$$

It is clear that F is differentiable and that the restriction $F|_{I \times \{(0,0)\}} = \alpha$. Moreover,

$$\det(dF_q) = \begin{vmatrix} x'(q) & 0 & 0 \\ y'(q) & 1 & 0 \\ z'(q) & 0 & 1 \end{vmatrix} = x'(q) \neq 0.$$

Thus, we can apply the inverse function theorem, which guarantees the existence of a neighborhood M of $\alpha(q)$ in $\mathbb{R}^3$ such that F^{-1} exists and is differentiable in M .

By the continuity of β , there exists a neighborhood N of r in J such that $\beta(N) \subseteq M$. When restricted to N , $h|_N = F^{-1} \circ \beta|_N$ is a composition of differentiable maps, so it is differentiable at r . Since r is arbitrary, h is differentiable on $\beta^{-1}(W)$.

The same argument applies to show h^{-1} is differentiable, so h is a diffeomorphism.

(b) Writing $\beta = \alpha \circ h$, we have

$$\int_{\tau_0}^{\tau} |\beta'(\tau)| d\tau = \int_{\tau_0}^{\tau} |(\alpha \circ h)'(\tau)| d\tau = \int_{\tau_0}^{\tau} |\alpha'(h(\tau))| \cdot |h'(\tau)| d\tau.$$

Since h is differentiable with differentiable inverse, h is continuous and h' can never be zero, meaning h is either always positive or always negative. This lets us get rid of the absolute value around $h'(\tau)$ to compute

$$\left| \int_{\tau_0}^{\tau} |\alpha'(h(\tau))| \cdot |h'(\tau)| d\tau \right| = \left| \int_{\tau_0}^{\tau} |\alpha'(h(\tau))| \cdot h'(\tau) d\tau \right| = \left| \int_{t_0}^t |\alpha'(t)| dt \right|$$

by u -substitution. □

Problem. 2.4.3. Show that the equation of the tangent plane of a surface which is the graph of a differentiable function $z = f(x, y)$, at the point $p_0 = (x_0, y_0)$, is given by

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

Solution. The surface admits a parametrization $\mathbf{x}: \mathbb{R}^2 \rightarrow S$ given by

$$\mathbf{x}(x, y) = (x, y, f(x, y)).$$

As per Proposition 1, the set of tangent vectors to S at $(x_0, y_0, f(x_0, y_0))$ is the image of the derivative

$$d\mathbf{x}_{p_0} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ f_x(x_0, y_0) & f_y(x_0, y_0) \end{pmatrix}$$

In particular a normal vector to the tangent plane is

$$\begin{pmatrix} 1 \\ 0 \\ f_x(x_0, y_0) \end{pmatrix} \wedge \begin{pmatrix} 0 \\ 1 \\ f_y(x_0, y_0) \end{pmatrix} = \begin{pmatrix} -f_x(x_0, y_0) \\ -f_y(x_0, y_0) \\ 1 \end{pmatrix}$$

which tells us that the tangent plane passing through $(x_0, y_0, f(x_0, y_0))$ has equation

$$z - f(x_0, y_0) - f_x(x_0, y_0)(x - x_0) - f_y(x_0, y_0)(y - y_0) = 0,$$

which rearranges to the given equation. $\square$

Problem. 2.4.8. Prove that if $L: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a linear map and $S \subseteq \mathbb{R}^3$ is a regular surface invariant under L , i.e. $L(S) \subseteq S$, then the restriction $L|_S$ is a differentiable map and

$$dL_p(w) = L(w)$$

for $p \in S$, $w \in T_p(S)$.

Solution. It is obvious that $L|_S$ is a differentiable map because L , being a linear map, is differentiable as a function $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ and $dL_p = L$ for all $p \in \mathbb{R}^3$ (again, as a function $\mathbb{R}^3 \rightarrow \mathbb{R}^3$). For a given $w \in T_p(S)$, let $\alpha: (-\varepsilon, \varepsilon) \rightarrow S$ be a differentiable parametrized curve with $\alpha(0) = p$, $\alpha'(0) = w$. Then, by the chain rule,

$$dL_p(w) = (L \circ \alpha)'(0) = dL_{\alpha(0)} \cdot \alpha'(0) = L(w)$$

where the multiplication $dL_{\alpha(0)} \cdot \alpha'(0)$ is matrix multiplication. $\square$

Problem. 2.4.15. Show that if all normals to a connected surface pass through a fixed point, the surface is contained in a sphere.

Solution. Let S be the surface and suppose WLOG (say after a translation) that the fixed point is the origin.

Claim — For any $p \in S$, if $V \subseteq S$ is the image of an open ball in $\mathbb{R}^2$ containing $\mathbf{x}^{-1}(p)$, then $\|p\| = \|x\|$ for every $x \in V$.

Proof. Let $U = \mathbf{x}^{-1}(V)$ be the open ball mentioned in the claim. For any $x \in V$, let $\alpha: I = (-\varepsilon, 1 + \varepsilon) \rightarrow \mathbb{R}^2$ be the function $\alpha(t) = (1 - t)q + tw$ for $q := \mathbf{x}^{-1}(p)$ and $w := \mathbf{x}^{-1}(x)$, where we choose ε small enough for $\alpha(I) \subseteq U$ to hold (such an ε exists because U we can find open balls around q and w that are contained in U). Notably, we are making good use of the convexity of U .

Then, α determines a regular curve through p and x on the surface S via the composed map $I \rightarrow U \rightarrow S$ defined by

$$t \xrightarrow{\alpha} (1 - t)q + tw \xrightarrow{\mathbf{x}} \mathbf{x}((1 - t)q + tw),$$

which we can easily verify is a differentiable homeomorphism whose derivative is never zero (really we just need to check this for α , but that's trivial).

Let $\beta = \mathbf{x} \circ \alpha$ be this composed map. On the interval I , we then have

$$\beta'(t) \cdot \beta(t) = 0$$

for all t since $\beta(t)$ is by assumption normal to S and $\beta'(t)$ is a tangent vector; but this is the derivative of $\beta(t) \cdot \beta(t) = \|\beta(t)\|^2$, so $\|\beta(t)\|^2$ must be constant as t varies across I , hence $\|p\| = \|\beta(0)\| = \|\beta(1)\| = \|x\|$ for all $x \in V$. $\square$

Claim — For any two points a, b in a connected open subset U of an arbitrary metric space, there is a finite sequence of open balls $B_0, B_1, \dots, B_k$ such that $B_i \subseteq U$ for all i , $a \in B_1$, $b \in B_k$, and $B_{j-1} \cap B_j \neq \emptyset$ for all $j = 1, \dots, k$.

Proof. For a given point $a \in U$, we say that $b \in U$ is *a-reachable* if such a sequence of open balls (which we will refer to as a “chain from a to b ”) exists for a and b . We will show that $R = U$, i.e. every point of U is a -reachable.

For any $x \in R$ we have a chain from a to x , in which the terminal ball, say $B(y, r)$, contains x . Then, we can find an open ball centered at x , such as $B(x, r - B(y, r))$, contained in $B(y, r)$ — that is, for any $x \in R$, there is an open ball centered at x that is contained in R . Thus, R is open.

On the other hand, R is closed with respect to U . Indeed, if $x \in U$ is a limit point of R , then there is an open ball B centered at x that intersects U at some point $y \neq x$. Then we have a chain $B_0, \dots, B_n$ from a to y , so $B_0, \dots, B_n, B$ is a chain from a to x , hence $x \in R$. Thus, R is closed in U because it contains its limit points.

Since path connectedness implies open set connectedness (there are no nontrivial clopen sets in U), we can conclude that, because R is both closed and open, it is either empty or equal to U , but clearly R is nonempty because $a \in R$, hence $R = U$ as desired. $\square$

To finish, let a and b be two arbitrary points in S . Suppose for a moment that a and b are not negative scalar multiples of one another, i.e. there is no positive real λ such that $a = -\lambda b$. Then we can apply a rotation, sending S to S' , so that a and b both have positive z -coordinate.

Let S'_+ be the intersection of S' with $\{(x, y, z) : z > 0\}$. The projection $\pi|_{S'_+}$ of S'_+ onto the xy -plane must be injective, else there exists a point at which the normal vector does not pass through the origin because the tangent plane will somewhere be perpendicular to the xy -plane (at some point where the distance to the origin is locally maximized); this can only occur at points where S' intersects the xy -plane.

Thus undoing the projection is a parametrization $\pi^{-1} : \pi(S'_+) \rightarrow S'_+$. Now $\pi(a)$ and $\pi(b)$ can be connected by a chain of open balls contained in $\pi(S'_+)$, at which point applying the first claim to each open ball in the chain and using the fact that consecutive balls in the sequence intersect shows that $\|a\| = \|b\|$.

The only other consideration is when $a = -\lambda b$ for a positive real λ , whence the rotation to obtain S' doesn't exist. In this case we pick an arbitrary third point c on the surface that is not collinear with a and b (just pick c to be in some neighborhood $V \subseteq S$ of a) and then apply the logic of the previous paragraph to get $\|a\| = \|c\| = \|b\|$. $\square$

Problem. 2.4.18. Prove that if a regular surface S meets a plane P in a single point p , then this plane coincides with the tangent plane of S at p .

Solution. Apply a rigid motion so that p maps to the origin and P gets sent to the xy -plane. Then, on a small enough neighborhood $U \subseteq S$ of p , the z -coordinates of all $x \in U$ have the same sign, say WLOG positive (except for a z -coordinate of zero at p). For any differentiable parametrized curve $\alpha : (-\varepsilon, \varepsilon) \rightarrow S$ on S with $\alpha(0) = p$, the z component function thus attains a local minimum at 0, meaning $\alpha'(0)$ has a z -coordinate of zero, i.e. it lies in the xy -plane. This holds for every such α , so every tangent vector at p lies in P , meaning P is the tangent plane of S at p . $\square$